

Measure Theory

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Problem

Consider a random (infinite) string of 0's and 1's

$$(\ell_1, \dots, \ell_n, \dots) \quad \ell_n \in \{0, 1\} \forall n \in \mathbb{N}$$

What is the probability that somewhere along the string, I find 100 consecutive zeros?

Comment

The statement of the problem is unclear. Lebesgue's machinery will clarify this

Define

$$X = \{x = (\ell_1, \dots, \ell_n, \dots) \mid \ell_n \in \{0, 1\} \forall n \in \mathbb{N}\}$$

This is, in a natural way, a compact metric space. Use the "Hamming-style" distance:

For $x = (\ell_1, \dots, \ell_n, \dots)$ and $x' = (\ell'_1, \dots, \ell'_n, \dots)$

$$d(x, x') = \sum_{n \in \mathbb{N}} \frac{|\ell_n - \ell'_n|}{2^n}$$

Since this is a metric space, we can consider the topology:

$$\mathcal{T} = \{G \subseteq X \mid G \text{ is open}\} \subseteq 2^X = \mathcal{P}(X)$$

Can consider

$$\mathfrak{B} = \sigma\text{-Alg}(\mathcal{T}) \quad (\text{Borel sigma algebra})$$

Then a probability measure is

$$P : \mathcal{B} \longrightarrow [0, 1]$$

Problem 1.1 says:

Let

$$S = \{x = (\ell_1, \dots, \ell_n, \dots) \in X \mid \exists m \in \mathbb{N} \text{ with } \ell_m = \ell_{m+1} = \dots = \ell_{m+99} = 0\}$$

check that $S \in \mathcal{B}$, and compute $P(S)$

1 Algebras of Sets

Fix a non-empty set X .

Definition 1.1**Algebra of Sets**

$\mathfrak{A} \subseteq 2^X$ is an *algebra of sets* if

1. $\emptyset \in \mathfrak{A}$
2. If $A \in \mathfrak{A}$, then $X \setminus A \in \mathfrak{A}$
3. If $A, B \in \mathfrak{A}$, then $A \cap B \in \mathfrak{A}$

Proposition 1.2**Properties of algebras of sets**

Let $\mathfrak{A}$ be an algebra of sets on X . Then

1. $X \in \mathfrak{A}$
2. If $A, B \in \mathfrak{A}$, then $A \cup B \in \mathfrak{A}$
3. If $A, B \in \mathfrak{A}$, then $A \setminus B \in \mathfrak{A}$
4. If $\mathfrak{B} \subseteq \mathfrak{A}$ is finite, then $\bigcap \mathfrak{B} \in \mathfrak{A}$
5. If $\mathfrak{B} \subseteq \mathfrak{A}$ is finite, then $\bigcup \mathfrak{B} \in \mathfrak{A}$

Example 1.1

Obviously, 2^X and $\{\emptyset, X\}$ are algebras of sets on X

Lemma 1.3**Intersections of algebras**

Let $\mathfrak{F}$ be any collection of algebras of sets on X , then $\bigcap \mathfrak{F}$ is an algebra of sets on X .

Proof: First note if $\mathfrak{F} = \emptyset$, then $\bigcap \mathfrak{F} = 2^X$ which is an algebra of sets on X . We now proceed with checking the axioms of an algebra of sets on X .

1. For every $\mathfrak{A} \in \mathfrak{F}$, we have $\emptyset \in \mathfrak{A}$, thus $\emptyset \in \bigcap \mathfrak{F}$.
2. If $A \in \bigcap \mathfrak{F}$, then $A \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$. Thus $X \setminus A \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$, and thus $X \setminus A \in \bigcap \mathfrak{F}$.
3. If $A, B \in \bigcap \mathfrak{F}$, then $A, B \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$. Thus $A \cap B \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$, and thus $A \cap B \in \bigcap \mathfrak{F}$.

□

Proposition 1.4**Algebra generated by a collection**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X , then there exists a unique algebra of sets $\mathfrak{A}$ on X with the following properties:

1. $\mathfrak{B} \subseteq \mathfrak{A}$
2. Whenever $\mathfrak{C}$ is an algebra of sets on X with $\mathfrak{B} \subseteq \mathfrak{C}$, then we have $\mathfrak{A} \subseteq \mathfrak{C}$

Proof: Define $\mathfrak{F}$ to be the collection of all algebras of sets on X that contain $\mathfrak{B}$. Then $\bigcap \mathfrak{F}$ is an algebra of sets by [Lemma 1.3](#) and satisfies (1) and (2). Uniqueness follows from (2). □

Definition 1.5**Algebra of sets generated by a collection of subsets**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X . We define the *algebra of sets generated by $\mathfrak{B}$* to be the unique algebra from [Proposition 1.4](#). We notate it as $\text{Alg}(\mathfrak{B})$.

1.1 Additive set-function on an algebra of sets

Definition 1.6

Additive set-function

Let $\mathfrak{A} \subseteq 2^X$ be an algebra of sets on X . A function $\mu : \mathfrak{A} \rightarrow [0, \infty]$ is called an *additive set-function on $\mathfrak{A}$* if

1. If $A, B \in \mathfrak{A}$ with $A \cap B = \emptyset$, then $\mu(A \cup B) = \mu(A) + \mu(B)$
2. $\mu(\emptyset) = 0$

Remark 1.7

The arithmetic of $[0, \infty]$ is standard, except for $0 \cdot \infty = \infty \cdot 0 = 0$, which is a useful convention in measure theory.

Remark 1.8

We almost don't need to require (2), it's just when μ is identically ∞ do things go wrong without (2)

Proposition 1.9

Properties of an additive set function

Let $\mathfrak{A}$ be an algebra of sets on X , and let μ be an additive set function on $\mathfrak{A}$. Then

1. If $A, C \in \mathfrak{A}$ with $A \subseteq C$, then $\mu(A) \leq \mu(C)$
2. If $A, B \in \mathfrak{A}$, then $\mu(A \cup B) \leq \mu(A) + \mu(B)$ (sub additivity)
3. If $A_1, \dots, A_n \in \mathfrak{A}$ are all pairwise disjoint, then $\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n)$
4. If $A_1, \dots, A_n \in \mathfrak{A}$, then $\mu(A_1 \cup \dots \cup A_n) \leq \mu(A_1) + \dots + \mu(A_n)$

2 Sigma Algebras

Definition 2.1

Sigma Algebra

$\mathfrak{M} \subseteq 2^X$ is called a *sigma algebra* on X if

1. $\emptyset \in \mathfrak{M}$
2. If $A \in \mathfrak{M}$, then $X \setminus A \in \mathfrak{M}$
3. If $(A_n)_{n=1}^\infty \subseteq \mathfrak{M}$, then $\bigcap_{n=1}^\infty A_n \in \mathfrak{M}$

Remark 2.2

countable unions work too, and so do finite operations

Lemma 2.3

Intersections of sigma algebras

Let $\mathfrak{F}$ be any collection of sigma algebras on X , then $\bigcap \mathfrak{F}$ is a sigma algebra on X .

Proof: basically identical to the proof for algebras of sets □

Proposition 2.4**Sigma algebra generated by a collection**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X , then there exists a unique sigma algebra $\mathfrak{A}$ on X with the following properties:

1. $\mathfrak{B} \subseteq \mathfrak{A}$
2. Whenever $\mathfrak{C}$ is a sigma algebra on X with $\mathfrak{B} \subseteq \mathfrak{C}$, then we have $\mathfrak{A} \subseteq \mathfrak{C}$

Proof: basically identical to the proof for algebras of sets □

Definition 2.5**Sigma algebra generated by a collection of subsets**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X . We define the *sigma algebra generated by $\mathfrak{B}$* to be the unique sigma algebra from [Proposition 2.4](#). We notate it as $\sigma\text{-Alg}(\mathfrak{B})$.

Definition 2.6**Borel Sigma Algebra**

Given a topological space $(X, \mathcal{T})$, the sigma algebra generated by $\mathcal{T}$ is called the *Borel sigma algebra on X* and written $\mathcal{B}_X := \sigma\text{-Alg}(\mathcal{T})$

2.1 Positive measure on a sigma algebra**Remark 2.7****Increasing limits and series in $[0, \infty]$**

1. Let $(s_n)_{n=1}^\infty$ be an *increasing* sequence in $[0, \infty]$ in the sense that

$$s_1 \leq s_2 \leq \dots$$

Such a sequence is sure to approach a limit $L \in [0, \infty]$. With a slight abuse of notation we can also write the formula $L = \sup\{s_n \mid n \in \mathbb{N}\}$ meaning that L is the smallest upper bound for all the s_n 's. This is usually written in connection to a bounded increasing sequence of numbers in $[0, \infty)$ – our “slight abuse of notation” is that we are also allowing it to cover the case $L = \infty$, which can occur when some of the s_n 's are equal to ∞ , or when $(s_n)_{n=1}^\infty$ is an unbounded sequence in $[0, \infty)$.

2. Let $(t_n)_{n=1}^\infty$ be any sequence in $[0, \infty]$. We can consider the finite sums

$$s_n := \sum_{i=1}^n t_i$$

which forms an increasing sequence in $[0, \infty]$. We can consider $L = \lim_{n \rightarrow \infty} s_n$ as in (1.), and use the familiar notation $L = \sum_{n=1}^\infty t_n$

Definition 2.8**Positive measure**

Let $\mathfrak{M} \subseteq 2^X$ be sigma algebra of subsets of X . We say $\mu : \mathfrak{M} \rightarrow [0, \infty]$ is a *positive measure* on $\mathfrak{M}$ if it satisfies:

1. $\mu(\emptyset) = 0$
- 2.

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

Whenever $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$ with the property that if $i, j \in \mathbb{N}$ with $i \neq j$ we have $A_i \cap A_j = \emptyset$

Remark 2.9

A positive measure is, in particular, finitely additive. Thus we can use all the properties in [Proposition 1.9](#)

Definition 2.10**Nomenclature involving measures**

1. If $\mathfrak{M}$ is a sigma algebra on X , then the pair $(X, \mathfrak{M})$ is called a *measurable space*
2. If we also have a positive measure μ on $\mathfrak{M}$, then the triple $(X, \mathfrak{M}, \mu)$ is called a *measure space*
3. If $\mu(X) < \infty$, then we say μ is a *finite measure*
4. If $\mu(X) = 1$, then we say μ is a *probability measure* and that $(X, \mathfrak{M}, \mu)$ is a *probability space*
5. We say that μ is *sigma finite* if there exists a sequence of sets $(X_n)_{n=1}^{\infty}$ in $\mathfrak{M}$ with
 - a. $X_1 \subseteq X_2 \subseteq \dots$ (The X_n 's are said to form an increasing chain)
 - b. $\bigcup_{n=1}^{\infty} X_n = X$
 - c. $\mu(X_n) < \infty$ for all $n \in \mathbb{N}$

Proposition 2.11**Sigma sub-additivity**

Let $\mathfrak{M}$ be a sigma algebra on X , and let μ be a positive measure on $\mathfrak{M}$. If $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

Proposition 2.12**Continuity along chains**

Let $\mathfrak{M}$ be a sigma algebra on X , and let μ be a positive measure on $\mathfrak{M}$.

1. If $(A_n)_{n=1}^{\infty}$ is an increasing chain in $\mathfrak{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

2. If $(A_n)_{n=1}^{\infty}$ is a decreasing chain in $\mathfrak{M}$ with $\mu(A_1) < \infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

Example 2.1

Example of “atomic” μ based on weights of individual points $x \in X$. Suppose you have $w : X \rightarrow [0, \infty]$. Let $\mathfrak{M} = 2^X$, for every $A \subseteq X$, put

$$\mu(A) = \sup \left\{ \sum_{x \in F} w(x) \mid F \subseteq A \text{ finite} \right\}$$

Then $(X, \mathfrak{M}, \mu)$ is a measure space. For the special case that $w(x) = 1, \forall x \in X$ is called the *counting measure* on X . For the special case that $w(x_0) = 1$ for some $x_0 \in X$ and $w(x) = 0, \forall x \neq x_0$ is called the *Dirac measure concentrated at x_0*

2.2 A glimpse of the Lebesgue measure on $\mathbb{R}$ **Note**

1. $\mathcal{B}_{\mathbb{R}}$ can be generated from the half open intervals
2. If $\mu, \nu : \mathcal{B}_{\mathbb{R}} \rightarrow [0, \infty]$ are positive measures such that

$$\mu((a, b]) = \nu((a, b]) < \infty, \forall a < b \in \mathbb{R}$$

It follows that $\mu = \nu$

Then the *Lebesgue measure* $\lambda : \mathcal{B}_{\mathbb{R}} \rightarrow [0, \infty]$ is determined via with requirement that $\lambda((a, b]) = b - a, \forall a < b \in \mathbb{R}$.

Another approach: λ is uniquely determined by the requirement that $\lambda(B + t) = \lambda(B)$ for every $B \in \mathcal{B}_{\mathbb{R}}$ and $t \in \mathbb{R}$ with the normalization that $\lambda((0, 1]) = 1$.

Note

For the *existence* of λ , will invoke an extension theorem of Caratheodory (to be done later)

Note

Interesting fact: there is no positive measure $\mu : 2^{\mathbb{R}} \rightarrow [0, \infty]$ which is translation invariant and $\mu((0, 1]) = 1$. This shows, in particular that $\mathcal{B}_{\mathbb{R}} \neq 2^{\mathbb{R}}$

3 Measurable functions

Definition 3.1**Measurable function**

Let $(X, \mathfrak{M}), (Y, \mathfrak{N})$ be measurable spaces and $f : X \rightarrow Y$. Then f is called $\mathfrak{M}/\mathfrak{N}$ -measurable if $f^{-1}(N) \in \mathfrak{M}$ for every $N \in \mathfrak{N}$.

Remark 3.2

Pre-images behave nicely with respect to the usual set operations.

Proposition 3.3**Composition of measurable maps**

Compositions of measurable maps are measurable.

Proof: Obvious □

Proposition 3.4**Measurability criterion**

Let $(X, \mathfrak{M}), (Y, \mathfrak{N})$ be measurable spaces. $f : X \rightarrow Y$, and $\mathfrak{C} \subseteq \mathfrak{N}$ such that $\sigma\text{-Alg}(\mathfrak{C}) = \mathfrak{N}$ and such $f^{-1}(C) \in \mathfrak{M}$ for every $C \in \mathfrak{C}$. Then f is measurable.

Proof: Let $\mathfrak{Q} = \{Q \subseteq Y \mid f^{-1}(Q) \in \mathfrak{M}\} \subseteq 2^Y$

Claim: $\mathfrak{Q}$ is a sigma algebra

Note by assumption $\mathfrak{C} \subseteq \mathfrak{Q}$ thus $\mathfrak{N} = \sigma\text{-Alg}(\mathfrak{C}) \subseteq \mathfrak{Q}$. Thus f is measurable. □

Corollary 3.5**Continuous maps are measurable**

Let X, Y be topological spaces, $f : X \rightarrow Y$ continuous. Then f is $\mathcal{B}_X/\mathcal{B}_Y$ -measurable.

3.1 The space of functions $\text{Bor}(X, \mathbb{R})$

Definition 3.6**Borel measurable real-valued functions**

Given a measurable space $(X, \mathfrak{M})$. Define

$$\text{Bor}(X, \mathbb{R}) := \{f : X \rightarrow \mathbb{R} \mid f \text{ is } \mathfrak{M}/\mathcal{B}_{\mathbb{R}}\text{-measurable}\}$$

Proposition 3.7**Algebra and lattice operations**

Let $(X, \mathfrak{M})$ be a measurable space with $f, g \in \text{Bor}(X, \mathbb{R})$. Then

1. $f + g \in \text{Bor}(X, \mathbb{R})$
2. For any $c \in \mathbb{R}$, $cf \in \text{Bor}(X, \mathbb{R})$
3. $f \cdot g \in \text{Bor}(X, \mathbb{R})$
4. $f \vee g, f \wedge g \in \text{Bor}(X, \mathbb{R})$

Proof of 3: Let $f, g \in \text{Bor}(X, \mathbb{R})$. Let $h : X \rightarrow \mathbb{R}^2$ given by $h(x) = (f(x), g(x))$. Let $p : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $p(s, t) = s \cdot t$.

Claim: h is measurable

Claim: p is measurable

Claim: $f \cdot g = p \circ h$

Thus $f \cdot g \in \text{Bor}(X, \mathbb{R})$ □

Exercise 3.1

Let $\mathfrak{S}$ be the collection of open squares in $\mathbb{R}^2$. Then $\sigma\text{-Alg}(\mathfrak{S}) = \mathcal{B}_{\mathbb{R}^2}$.

Remark 3.8

Let $(X, \mathfrak{M})$ be a measurable space. Then $\text{Bor}(X, \mathbb{R})$ is an *algebra of functions* and a *lattice of functions*.

Remark 3.9

Operations with only one function. If $f \in \text{Bor}(X, \mathbb{R})$, then $|f| \in \text{Bor}(X, \mathbb{R})$. Use algebraic stability properties to prove this.

Example 3.1**Problem 3.1**

Claim: Let $\mathfrak{C} = \{(-\infty, t], \mid t \in \mathbb{R}\} \subseteq \mathcal{B}_{\mathbb{R}}$. Then $\sigma\text{-Alg}(\mathfrak{C}) = \mathcal{B}_{\mathbb{R}}$.

Proof: Denote $\sigma\text{-Alg}(\mathfrak{C}) = \mathfrak{P}$. Clearly, $\mathfrak{P} \subseteq \mathcal{B}_{\mathbb{R}}$. If we can show that $G \in \mathfrak{P}$ for all $G \subseteq \mathbb{R}$ open, then we will be done because we will have that $\mathfrak{P}$ is a sigma-algebra and that $\mathfrak{P} \supseteq \mathcal{T}_{\mathbb{R}}$.

Now, every $\emptyset \neq G \subseteq \mathbb{R}$ open can be written as a countable union of open intervals. So then, it suffices to check that $\mathfrak{P}$ contains all of the open intervals of $\mathbb{R}$. This is left as an exercise. □

Let $(X, \mathfrak{M})$ be a measurable space, and suppose $f : X \rightarrow \mathbb{R}$ satisfies $f^{-1}((-\infty, t]) \in \mathfrak{M}$ for all $t \in \mathbb{R}$. That is, $f^{-1}(C) \in \mathfrak{M}$ for all $C \in \mathfrak{C}$. Then by [Proposition 3.4](#), $f^{-1}(B) \in \mathfrak{M}$ for all $B \in \mathcal{B}_{\mathbb{R}}$, hence $f \in \text{Bor}(X, \mathbb{R})$.

4 Convergence properties of $\text{Bor}(X, \mathbb{R})$

Proposition 4.1

Closure under suprema and infima

Let $(X, \mathfrak{M})$ be a measurable space.

1. Let $f : X \rightarrow \mathbb{R}$ be a function. Let $(f_n)_{n=1}^{\infty}$ be a sequence of functions in $\text{Bor}(X, \mathbb{R})$ such that

$$\sup\{f_n(x) \mid n \in \mathbb{N}\} = f(x)$$

for all $x \in X$. Then it follows that $f \in \text{Bor}(X, \mathbb{R})$.

2. Let $g : X \rightarrow \mathbb{R}$ be a function. Let $(g_n)_{n=1}^{\infty}$ be a sequence of functions in $\text{Bor}(X, \mathbb{R})$ such that

$$\inf\{g_n(x) \mid n \in \mathbb{N}\} = g(x)$$

for all $x \in X$. Then it follows that $g \in \text{Bor}(X, \mathbb{R})$.

Proof:

1. It suffices to show that $f^{-1}((-\infty, t]) \in \mathfrak{M}$ for all $t \in \mathbb{R}$. This is left as an exercise.
2. Exercise.

□

Remark 4.2

Let $(t_n)_{n=1}^{\infty}$ be a bounded sequence in $\mathbb{R}$. Then there exists $\ell_+ \in \mathbb{R}$ such that

1. There exists a subsequence of $(t_n)_{n=1}^{\infty}$ which converges to ℓ_+ .
2. There is no subsequence of $(t_n)_{n=1}^{\infty}$ which converges to an $\ell > \ell_+$.

Define

$$\limsup_{n \rightarrow \infty} t_n := \ell_+$$

We can obtain ℓ_+ as

$$\ell_+ = \lim_{k \rightarrow \infty} \left(\sup_{n \geq k} t_n \right) = \inf_{k \geq 1} \left(\sup_{n \geq k} t_n \right)$$

We can define $\liminf_{n \rightarrow \infty} t_n$ similarly.

Fix a measurable space $(X, \mathfrak{M})$

Proposition 4.3

1. Let $f : X \rightarrow \mathbb{R}$. Suppose we could find $f_1, f_2, \dots \in \text{Bor}(X, \mathbb{R})$ such that for every $x \in X$, the sequence $(f_n(x))_{n=1}^\infty$ is bounded with

$$\limsup_{n \rightarrow \infty} f_n(x) = f(x)$$

Then it follows that $f \in \text{Bor}(X, \mathbb{R})$

2. Let $g : X \rightarrow \mathbb{R}$. Suppose we could find $g_1, g_2, \dots \in \text{Bor}(X, \mathbb{R})$ such that for every $x \in X$, the sequence $(g_n(x))_{n=1}^\infty$ is bounded with

$$\liminf_{n \rightarrow \infty} g_n(x) = g(x)$$

Then it follows that $g \in \text{Bor}(X, \mathbb{R})$

Proof of 1: For every $k \in \mathbb{N}$, define the helper function $h_k : X \rightarrow \mathbb{R}$ by

$$h_k(x) = \sup \underbrace{\{f_n(x) \mid n \geq k\}}_{\text{bounded by hypothesis}}, \quad \forall x \in X$$

Observe $f_k, f_{k+1}, \dots \in \text{Bor}(X, \mathbb{R})$. We can invoke [Proposition 4.1](#), to conclude that $h_k \in \text{Bor}(X, \mathbb{R})$. Observe for every $x \in X$, we have

$$h_1(x) \geq h_2(x) \geq \dots$$

with

$$f(x) = \lim_{k \rightarrow \infty} h_k(x) = \inf\{h_k(x) \mid k \in \mathbb{N}\}$$

This is because $f(x) = \limsup_{n \rightarrow \infty} f_n(x)$ by hypothesis. But then

$$h_k \in \text{Bor}(X, \mathbb{R}), \quad \forall k \in \mathbb{N}$$

and

$$f(x) = \inf\{h_k(x) \mid k \in \mathbb{N}\} \quad \forall x \in X$$

thus $f \in \text{Bor}(X, \mathbb{R})$ by [Proposition 4.1](#). □

Proof of 2: We can reduce to (1), by introducing $f : X \rightarrow \mathbb{R}$, $f(x) = -g(x) \quad \forall x \in X$ and introducing $f_n = -g_n \in \text{Bor}(X, \mathbb{R})$, $\forall n \in \mathbb{N}$, then apply (1) to $f(x) = \limsup_{n \rightarrow \infty} f_n(x)$, to get $f \in \text{Bor}(X, \mathbb{R})$, hence $g \in \text{Bor}(X, \mathbb{R})$. □

Corollary 4.4

Let $f : X \rightarrow \mathbb{R}$. Suppose we could find $(f_n)_{n=1}^\infty$ in $\text{Bor}(X, \mathbb{R})$ such that $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, $\forall x \in X$. Then it follows that $f \in \text{Bor}(X, \mathbb{R})$.

Proof: Use [Proposition 4.3](#). □

Exercise 4.1

Suppose that $(X, \mathfrak{M}) = (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and consider the function $f' : \mathbb{R} \rightarrow \mathbb{R}$ (f' may not be continuous).

Solution:

Idea is to produce $(f_n)_{n=1}^{\infty}$ in $\text{Bor}(\mathbb{R}, \mathbb{R})$ such that $\lim_{n \rightarrow \infty} f_n(x) = f'(x)$. If we find the f_n 's then $f' \in \text{Bor}(\mathbb{R}, \mathbb{R})$ due to [Corollary 4.4](#). How do we find the f_n 's? For $x \in \mathbb{R}$

$$f'(x) = \lim_{n \rightarrow \infty} \frac{f(x + \frac{1}{n}) - f(x)}{\frac{1}{n}} = \lim_{n \rightarrow \infty} n(u_n(x) - f(x))$$

With $u_n(x) = f(x + \frac{1}{n})$. So then, for every $n \in \mathbb{N}$, let $u_n : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $u_n(x) = f(x + \frac{1}{n})$, $\forall x \in \mathbb{R}$ and let $f_n = n(u_n - f)$. Every f_n is continuous, hence in $\text{Bor}(\mathbb{R}, \mathbb{R})$ by [Corollary 3.5](#) and $\lim_{n \rightarrow \infty} f_n(x) = f'(x)$ as we wanted.

4.1 Approximation by simple functions**Definition 4.5****Simple functions**

A function $f \in \text{Bor}(X, \mathbb{R})$ is said to be *simple* when it takes finitely many values, that is $f(X)$ is a finite subset of $\mathbb{R}$. We denote

$$\text{Bor}_s(X, \mathbb{R}) := \{f \in \text{Bor}(X, \mathbb{R}) \mid f \text{ is simple}\}$$

We also introduce the spaces

$$\text{Bor}^+(X, \mathbb{R}) = \{f \in \text{Bor}(X, \mathbb{R}) \mid f(x) \geq 0 \ \forall x \in X\}$$

and

$$\text{Bor}_s^+(X, \mathbb{R}) := \text{Bor}_s(X, \mathbb{R}) \cap \text{Bor}^+(X, \mathbb{R})$$

Note

For $A \in \mathfrak{M}$, we have $\chi_A \in \text{Bor}_s(X, \mathbb{R})$, where $\chi_A : X \rightarrow \mathbb{R}$ is defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases}$$

called the *indicator function* of A . Also note that χ_A is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$ -measurable since for every $S \in \mathcal{B}_{\mathbb{R}}$ we have that

$$\chi_A^{-1}(S) \in \{\emptyset, A, X \setminus A, X\} \subseteq \mathfrak{M}$$

Note

It can be shown that

$$\text{Bor}_s(X, \mathbb{R}) = \underbrace{\text{span}\{\chi_A \mid A \in \mathfrak{M}\}}_{\substack{\text{set of all linear combinations} \\ \text{of indicator functions} \\ \chi_A \text{ with } A \in \mathfrak{M}}}$$

Remark 4.6**Binning values of a non-negative Borel function**

Let $f \in \text{Bor}^+(X, \mathbb{R})$. For every $n \in \mathbb{N}$, let $f_n : X \rightarrow \mathbb{R}$ be the n -th binning of the values of f , defined as follows. For each $x \in X$:

1. If $f(x) \geq n$, define $f_n(x) := n$.
2. If $f(x) < n$, pick the unique $k \in \{1, \dots, n \cdot 2^n\}$ such that

$$\frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}$$

and define $f_n(x) := \frac{k-1}{2^n}$.

Proposition 4.7**Binning approximation**

Let $f \in \text{Bor}^+(X, \mathbb{R})$, and for every $n \in \mathbb{N}$, let f_n be the n -th binning of the values of f . Then

1. $f_n \in \text{Bor}_s^+(X, \mathbb{R})$ for every $n \in \mathbb{N}$
2. $f_n \leq f_{n+1}$ for every $n \in \mathbb{N}$
3. For every $x \in X$,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

5 Integration of functions in $\text{Bor}^+(X, \mathbb{R})$

5.1 The map $L_s^+ : \text{Bor}_s^+(X, \mathbb{R}) \rightarrow [0, \infty]$

Fix a measure space $(X, \mathfrak{M}, \mu)$

Note

For didactical reasons, it is convenient to first define $\int_X f(x) d\mu(x)$ for $f \in \text{Bor}^+(X, \mathbb{R})$. In this case, the integral is always defined but may be ∞ . For general $f \in \text{Bor}(X, \mathbb{R})$, the integral may not be defined; to get a finite integral, we need $f \in \mathcal{L}^1(\mu)$.

Moreover, it will be convenient to introduce the integral on $\text{Bor}^+(X, \mathbb{R})$ as a map

$$L^+ : \text{Bor}^+(X, \mathbb{R}) \rightarrow [0, \infty].$$

We will get L^+ in two parts:

1. Define $L_s^+ : \text{Bor}_s^+(X, \mathbb{R}) \rightarrow [0, \infty]$
2. Extend L_s^+ to a map $L^+ : \text{Bor}^+(X, \mathbb{R}) \rightarrow [0, \infty]$

The map $L_s^+ : \text{Bor}_s^+(X, \mathbb{R}) \rightarrow [0, \infty]$

Remark 5.1**Easily verified properties of $\text{Bor}_s^+(X, \mathbb{R})$**

1. If $f, g \in \text{Bor}_s^+(X, \mathbb{R})$, then $f + g \in \text{Bor}_s^+(X, \mathbb{R})$
2. If $f \in \text{Bor}_s^+(X, \mathbb{R})$ and $\alpha \in [0, \infty)$, then $\alpha f \in \text{Bor}_s^+(X, \mathbb{R})$
3. If $A \in \mathfrak{M}$, then $\chi_A \in \text{Bor}_s^+(X, \mathbb{R})$

Note

As a consequence of (3), taking $A = \emptyset$ and $A = X$, we get $\underline{1}, \underline{0} : X \rightarrow \mathbb{R} \in \text{Bor}_s^+(X, \mathbb{R})$.

Theorem 5.2**Integral of non-negative simple functions**

There exists a map

$$L_s^+ : \text{Bor}_s^+(X, \mathbb{R}) \longrightarrow [0, \infty],$$

uniquely determined, such that the following conditions are satisfied:

(i) Additivity:

$$L_s^+(f + g) = L_s^+(f) + L_s^+(g)$$

for all $f, g \in \text{Bor}_s^+(X, \mathbb{R})$

(ii) Positive homogeneity:

$$L_s^+(\alpha f) = \alpha L_s^+(f)$$

for all $f \in \text{Bor}_s^+(X, \mathbb{R})$ and $\alpha \in [0, \infty)$

(iii) Relation to the measure:

$$L_s^+(\chi_A) = \mu(A)$$

for all $A \in \mathfrak{M}$

In order to prove the existence of the map L_s^+ , we will take advantage of the fact that one can describe the functions from $\text{Bor}_s^+(X, \mathbb{R})$ in terms of measurable divisions of X .

Definition 5.3**Measurable division and intersections of measurable divisions**

1. A *measurable division* of X is a set $\{A_1, \dots, A_n\} \subseteq \mathfrak{M}$ of pairwise disjoint, non-empty sets such that $\bigcup_{i=1}^n A_i = X$
2. Given two measurable divisions $\Delta = \{A_1, \dots, A_n\}$ and $\Gamma = \{B_1, \dots, B_m\}$ of X , one can “intersect” them, to form a new measurable division of X , denoted $\Delta \wedge \Gamma$, defined by

$$\Delta \wedge \Gamma := \{A_i \cap B_j \mid 1 \leq i \leq n, 1 \leq j \leq m, \text{ and } A_i \cap B_j \neq \emptyset\}$$

Lemma 5.4**Simple functions and measurable divisions**

Let $f \in \text{Bor}_s^+(X, \mathbb{R})$.

1. There exists a measurable division $\Delta = \{A_1, \dots, A_n\}$ of X and numbers $\alpha_1, \dots, \alpha_n \in [0, \infty)$ such that

$$f = \alpha_1 \chi_{A_1} + \dots + \alpha_n \chi_{A_n}$$

2. Suppose that $\Delta = \{A_1, \dots, A_n\}$ and $\alpha_1, \dots, \alpha_n \in [0, \infty)$ are as in (1.), and suppose that $\Gamma = \{B_1, \dots, B_m\}$ is a second measurable division of X . Then

$$f = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \alpha_i \chi_{A_i \cap B_j}$$

Proof of 2: For every $1 \leq i \leq n$, write

$$\chi_{A_i} = \chi_{A_i} \cdot 1 = \chi_{A_i} (\chi_{B_1} + \dots + \chi_{B_m}) = \sum_{j=1}^m \chi_{A_i \cap B_j} = \sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \chi_{A_i \cap B_j}$$

Then

$$f = \sum_{i=1}^n \alpha_i \chi_{A_i} = \sum_{i=1}^n \alpha_i \left(\sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \chi_{A_i \cap B_j} \right) = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \alpha_i \chi_{A_i \cap B_j}$$

as desired. □

Lemma 5.5**Well-definedness of the simple integral formula**

Let $f \in \text{Bor}_s^+(X, \mathbb{R})$. Suppose we have two ways of writing f as a linear combination:

$$f = \alpha_1 \chi_{A_1} + \dots + \alpha_n \chi_{A_n}$$

and

$$f = \beta_1 \chi_{B_1} + \dots + \beta_m \chi_{B_m},$$

where $\Delta = \{A_1, \dots, A_n\}$ and $\Gamma = \{B_1, \dots, B_m\}$ are measurable divisions of X , and $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m \in [0, \infty)$. Then

$$\sum_{i=1}^n \alpha_i \mu(A_i) = \sum_{j=1}^m \beta_j \mu(B_j) \in [0, \infty].$$

Proof: For every $1 \leq i \leq n$, let us write

$$A_i = A_i \cap X = A_i \cap (B_1 \cup \dots \cup B_m) = \bigcup_{j=1}^m (A_i \cap B_j) = \bigcup_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} (A_i \cap B_j)$$

Since the union is disjoint,

$$\mu(A_i) = \sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \mu(A_i \cap B_j)$$

so

$$\sum_{i=1}^n \alpha_i \mu(A_i) = \sum_{i=1}^n \alpha_i \sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \mu(A_i \cap B_j) = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \alpha_i \mu(A_i \cap B_j)$$

Similarly,

$$\sum_{j=1}^m \beta_j \mu(B_j) = \sum_{j=1}^m \beta_j \sum_{\substack{1 \leq i \leq n, \\ B_j \cap A_i \neq \emptyset}} \mu(B_j \cap A_i) = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \emptyset}} \beta_j \mu(A_i \cap B_j)$$

Claim: If $A_i \cap B_j \neq \emptyset$, then $\alpha_i = \beta_j$

Observe that this claim will finish the proof. To verify the claim, suppose $A_i \cap B_j \neq \emptyset$. Pick $x \in A_i \cap B_j$, and evaluate $f(x)$ in two ways:

$$f(x) = \alpha_1 \chi_{A_1}(x) + \dots + \alpha_n \chi_{A_n}(x) = \alpha_i$$

similarly

$$f(x) = \beta_1 \chi_{B_1}(x) + \dots + \beta_m \chi_{B_m}(x) = \beta_j,$$

so $\alpha_i = f(x) = \beta_j$ □

Proof of existence of L_s^+ from Theorem 5.2: For every $f \in \text{Bor}_s^+(X, \mathbb{R})$, we define the quantity $L_s^+(f) \in [0, \infty]$ by the following procedure:

Consider a writing

$$f = \alpha_1 \chi_{A_1} + \dots + \alpha_n \chi_{A_n}$$

with $\Delta = \{A_1, \dots, A_n\}$ a measurable division of X and $\alpha_1, \dots, \alpha_n \in [0, \infty)$, (*)
and put

$$L_s^+(f) := \sum_{i=1}^n \alpha_i \mu(A_i) \in [0, \infty].$$

Lemma 5.5 assures us that the procedure (*) is coherent, in the sense that the proposed quantity $L_s^+(f)$ only depends on f and not on the choice of how we write f as a linear combination of indicators.

The map $L_s^+ : \text{Bor}_s^+(X, \mathbb{R}) \longrightarrow [0, \infty]$

With $(*)$, we have thus defined a map

$$L_s^+ : \text{Bor}_s^+(X, \mathbb{R}) \longrightarrow [0, \infty].$$

We must now check that this map has the properties (i), (ii), (iii) indicated in [Theorem 5.2](#). $\square$

5.2 The map $L^+ : \text{Bor}^+(X, \mathbb{R}) \longrightarrow [0, \infty]$

Definition 5.6

Integral of non-negative Borel functions

For every function $u \in \text{Bor}^+(X, \mathbb{R})$, we define

$$L^+(u) := \sup\{L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\} \in [0, \infty].$$

Remark 5.7

Comments on the formula for L^+

- We use letters such as $u, v, \dots$ for general functions in $\text{Bor}^+(X, \mathbb{R})$, reserving $f, g, \dots$ for functions in $\text{Bor}_s^+(X, \mathbb{R})$.
- In the formula above, $f \leq u$ means $f(x) \leq u(x)$ for every $x \in X$.
- The set over which we take the supremum is non-empty because it contains $\underline{0}$.
- The supremum is taken in $[0, \infty]$, so it is allowed to be ∞ .

Proposition 5.8

L^+ extends L_s^+

L^+ is an extension of L_s^+ . That is, if $u \in \text{Bor}_s^+(X, \mathbb{R})$, then the newly defined quantity $L^+(u)$ is the same as $L_s^+(u)$.

Proof: Must show

$$\sup\{L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\} = L_s^+(u)$$

“ $\geq$ ” holds because $L_s^+(u)$ is an element of $\{L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\}$. For “ $\leq$ ”, we must show $L_s^+(u)$ is an upper bound for $\{L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\}$. Indeed, if $f \in \text{Bor}_s^+(X, \mathbb{R})$ with $f \leq u$, then, using [Theorem 5.2](#),

$$L_s^+(u) = L_s^+(f + (u - f)) = L_s^+(f) + L_s^+(u - f) \geq L_s^+(f)$$

$\square$

Proposition 5.9

Monotonicity and homogeneity of L^+

1. L^+ is an increasing map: if $u, v \in \text{Bor}^+(X, \mathbb{R})$ with $u \leq v$, then $L^+(u) \leq L^+(v)$
2. L^+ is homogeneous: if $u \in \text{Bor}^+(X, \mathbb{R})$, $\alpha \in [0, \infty)$, then $L^+(\alpha u) = \alpha L^+(u)$

Proof of 1: Since $u \leq v$, we have

$$\{f \in \text{Bor}_s^+(X, \mathbb{R}) \mid f \leq u\} \subseteq \{f \in \text{Bor}_s^+(X, \mathbb{R}) \mid f \leq v\}.$$

Thus

The map $L^+ : \text{Bor}^+(X, \mathbb{R}) \longrightarrow [0, \infty]$

$$\begin{aligned}
L^+(u) &= \sup\{L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\} \\
&\leq \sup\{L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq v\} \\
&= L^+(v).
\end{aligned}$$

□

Proof of 2: If $\alpha = 0$, then $L^+(\alpha u) = 0 = \alpha L^+(u)$, where we use the convention $0 \cdot \infty = 0$.

Suppose $\alpha > 0$. Then

$$\{g \in \text{Bor}_s^+(X, \mathbb{R}) \mid g \leq \alpha u\} = \{\alpha f \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\}.$$

Thus,

$$\begin{aligned}
L^+(\alpha u) &= \sup\{L_s^+(\alpha f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\} \\
&= \sup\{\alpha L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\} \\
&= \alpha \sup\{L_s^+(f) \mid f \in \text{Bor}_s^+(X, \mathbb{R}), f \leq u\} \\
&= \alpha L^+(u),
\end{aligned}$$

where the second equality uses [Theorem 5.2](#).

□

Remark 5.10

Additivity and MCT

L^+ is additive: $L^+(u + v) = L^+(u) + L^+(v)$ for every $u, v \in \text{Bor}^+(X, \mathbb{R})$. We can show $L^+(u + v) \geq L^+(u) + L^+(v)$ by an easy argument using the definition, but the other inequality is not so easy. Instead we move to the Monotone Convergence Theorem (MCT), which will prove it. Note we have to make sure not to use additivity in the proof of MCT.

6 The Monotone Convergence Theorem

Fix a measure space $(X, \mathfrak{M}, \mu)$ from chapter 5. We will work with the map $L^+ : \text{Bor}^+(X, \mathbb{R}) \rightarrow [0, \infty]$.

Notation 6.1

Increasing convergence

For u and $u_1, u_2, \dots \in \text{Bor}^+(X, \mathbb{R})$, we will use the shorthand $u_n \nearrow u$ to mean that $u_1 \leq u_2 \leq \dots$ and $u_n \rightarrow u$ pointwise.

Theorem 6.2

Monotone Convergence Theorem

Let $u, u_1, u_2, \dots \in \text{Bor}^+(X, \mathbb{R})$ with $u_n \nearrow u$. Then

$$\lim_{n \rightarrow \infty} L^+(u_n) = L^+(u)$$

(an increasing limit in $[0, \infty]$)

Proposition 6.3

Additivity of L^+

L^+ is additive: $L^+(u + v) = L^+(u) + L^+(v)$ for every $u, v \in \text{Bor}^+(X, \mathbb{R})$